Prajwal Rao

Lafayette, Indiana, 47901, USA • +1 (612) 282-5351 • prajwalipj@gmail.com • LinkedIn • GitHub

Education

Masters in Electrical and Computer Engineering	**Purdue University**	*Aug 2022 - Aug 2024*
Bachelors in Electronics and Communication Engineering	**Ramaiah Institute of Technology**	*Aug 2013 - Jun 2017*

Work Experience

Purdue University - Graduate Research Assistant - *(Computer Vision for Precision Livestock Farming)* *Jan 2023 - Aug 2024*

- **Computer Vision:** Developed and deployed a real-time pipeline at Purdue Dairy, achieving 95% accuracy in estimating feed volume using a stereo camera system
- **3D Vision Simulation:** Created a high-accuracy (98.1%) barn environment prototype using Blender for system prototyping
- **Deep Learning:** Fine-tuned keypoint detection for bird phenotyping, applied to thousands of ducks, by developing an intuitive user interface

Gnani.ai - Senior AI Engineer - *(Conversational AI; Text-to-Speech (TTS); Video Lipsync)* *Jul 2020 – May 2022*

- **R&D:** Engineered a text-to-speech (TTS) system using customized versions of Tacotron, FastSpeech, and FastPitch architectures
- **Customer-Facing:** Boosted customer acquisition by 83% by implementing SSML support, including features such as break, pitch, speed, and phonetic representation
- **Product Offerings:** Expanded product capabilities and maintained continuous integration with multilingual support, incorporating English (US, IN), Hindi, Kannada, Marathi, Tamil, and Telugu, with both male and female voice options
- **MLOps:** Optimized pipeline for faster-than-real-time inferencing, achieving an average processing time of 50 ms per word and saving the company $1 million in cloud infrastructure costs
- **API Development:** Enhanced product deployment by building concurrent APIs using Httpx and gRPC frontends and increased customer satisfaction by 20%
- **Video GenAI:** Advanced the LipSync algorithm using scraped data, boosting visual quality from 480p to 1080p and overcoming challenges with low data fine-tuning to achieve superior performance
- **Multimodal Research:** Implemented voice cloning solutions using just 30 minutes of data, and developed proof of concept 3D Face reconstruction using FSGAN technologies to achieve high-quality outcomes

Indian Institute of Science - Technical Associate - *(Deep learning; Vision AI)* *Jun 2018 – Jun 2020*

- **Video Analytics:** Partnered with a team of researchers to create a traffic pattern analysis framework, which was successfully launched by the government for enhanced traffic monitoring
- **Model Quantization:** Accelerated YOLO-v3 performance using TensorRT, achieving a 10x increase in inference speed
- **Object Detection:** Improved object detection accuracy, increasing mean average precision (mAP) by 15%.
- **Inference Optimization:** Scaled the detection framework to handle parallel inferencing across 300 cameras, up from 1, significantly expanding its capacity and efficiency
- **API Integration:** Collaborated with cross-functional teams to design a speech interaction system for Hanson Robotics, incorporating wake-word detection, automatic speech recognition, and integration with the Sophia SDK to enhance functionality

Gnani.ai - AI Engineer - *(Conversational AI; Automatic Speech Recognition (ASR))* *Aug 2017 - Mar 2018*

- **Data Collection:** Designed and implemented a high-efficiency data collection and management pipeline, optimizing data preprocessing for system evaluation and failure analysis
- **Language Models:** Trained an automatic speech recognition (ASR) model to achieve 92% accuracy for the Kannada language, enhancing language processing capabilities
- **Pipeline Optimization:** Enabled support for over 1,000 concurrent streams by integrating a real-time multiprocessing decoder, significantly improving system performance
- **Machine Learning:** Enhanced Google's single keyword spotting network to recognize 350 keywords, achieving an accuracy of 83% and expanding the model's functionality

Skills

Speech domain: Speech recognition (ASR), Text-to-Speech (TTS), Voice Biometric, Replay Attack
Vision domain: Object detection, Image Classification, Video generation
Text domain: Transformers, GPT, LLM
Programming Skills and Tools: Python (OOPs, PyTorch, Tensorflow, TorchServe, TensorRT, OpenCV, Numpy, Pandas), Shell Scripting (ssh, bash, vim), LaTeX | ***Analytics and Visualization:*** Plotly, MatplotLib, Streamlit | ***Cloud and Data:*** AWS Storage, MongoDB
Other Tools: Docker, Git, Linux, Blender, Kaldi, ESPNet, Onnx, Emacs, Jupyter

Publications

- **Patents**
  - Method for Facilitating Speech Activity Detection for Streaming Speech Recognition [US20220358913A1]
- **Papers**
  - Real-Time Cattle Intake Monitoring Using Computer Vision - Electronic Imaging 2024 (Guide: Prof. Amy Reibman; Jan'24)
  - Link Speed Estimation for Traffic Flow Modelling Based on Video Feeds from Monocular Cameras - Intelligent Transportation Systems Society, 2020 (Guide: Prof. Raghu Krishnapuram; Jan'19 - Dec'19)
  - Classification of Lung Cancer using Deep Convolutional Neural Networks - 2nd International Conference on Contemporary Computing and Informatics, 2016 (Guide: Prof. Raghuram Srinivasan; July'16 – Dec'16)
  - Character Recognition using Artificial Bee-Colony Algorithm - 14th IEEE India Council International Conference, 2017 (Guide: Prof. KG Srinivasa; July'17 - Dec'17)

Leadership

***Speech Team* - Gnani - Project Ownership:** Individual responsibility from ideation to production for video analytics, TTS, ASR projects
***Speech Team Lead* - Gnani - Mentorship:** Mentored team of 4 in speech domain. Coached 3 full time engineers for Speech Team
***Patent Team Lead* - Gnani - Guidance:** Guided contributors in submitting patents. Collectively submitted 4 patents in 1 year
***Guest Lecture* - Ramaiah Institute of Technology** – *"Introduction to Image Classification and Object Detection"* – Aug'19